

Visualization Methods for Big and High-Dimensional Acoustic Data

Achilleas Xydis

ETH Zurich

Chaoyu Du

ETH Zurich

Romana Rust

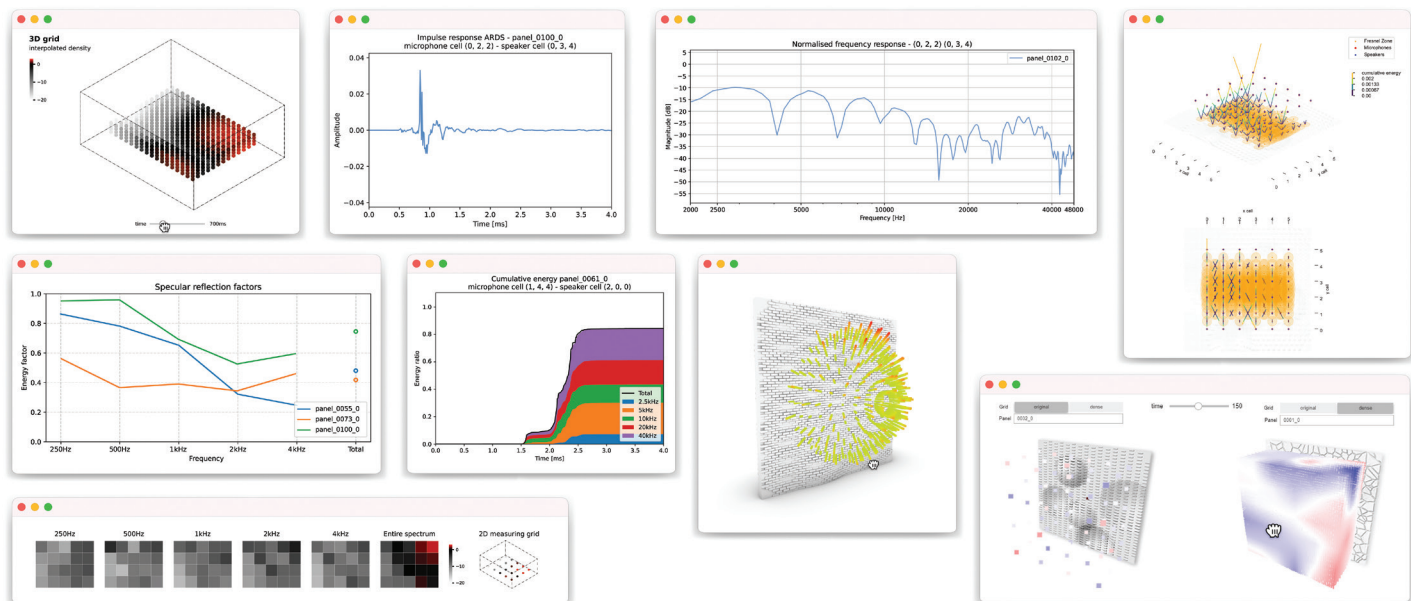
ETH Zurich

Fabio Gramazio

ETH Zurich

Matthias Kohler

ETH Zurich



1

ABSTRACT

Acoustics are rarely included in architectural design because available acoustic analysis tools are cumbersome and require expert knowledge in acoustics. This exclusion from the design phase could lead to late-stage design modifications, potential delays, and increased building costs. On the contrary, their inclusion can improve the acoustic properties of spaces and ensure a seamless design integration. This can be achieved by providing architects with easy-to-use visualization tools to study the relationship between geometry and sound without expert knowledge in acoustics. Available acoustic datasets can enable the development of such visualizations, but recent technological advances have increased their complexity and size. Although existing data science methods can process and analyze them, it remains challenging to develop easy-to-use and informative visualizations for architects and non-acoustic experts.

This research proposes a novel approach for interactive visualizations of acoustic datasets for architects and non-acoustic experts. It introduces a series of simple acoustic properties for users with basic knowledge of acoustics and describes methods for low- and high-dimensional data visualizations. It describes the computational workflow and uses a design scenario to demonstrate the proposed visualizations. Finally, it discusses the challenges of developing such methods, their advantages, limitations, and future work.

1 A selection of low and high-dimensional visualizations for acoustic data available in the presented computational workflow

INTRODUCTION

Sound visualizations are an integral part of the study of room acoustics. Since the introduction of Ultrasonic-Schlieren photography by Wallace C. Sabine in 1912 (Sabine 1922), acousticians have developed multiple methods to visualize the invisible-to-the-naked-eye propagation of sound and its interaction with surfaces. Advancements in measurement techniques and computer simulations make it easier today to measure and simulate sound propagation. However, our display mediums are still two-dimensional, and visualization tools face the challenge of projecting high-dimensional data on flat screens or paper. This challenge becomes apparent when we try to display even the most essential acoustical features (frequencies, amplitude, phase, direction) simultaneously, making the visualization unreadable or overwhelming. Looking at recent developments in acoustic measurement technology and advances in computing power and storage capabilities, acoustic data collection has the tendency to increase in complexity and size rather than simplify. Large and often heterogeneous data sets make it challenging to develop informative visualizations (Genender-Feltheimer 2018) because such datasets have surpassed human cognitive capabilities when explored through simple data analysis tools (Van Long and Linsen 2011). Therefore, the challenge of visualizing large acoustic datasets to make them accessible and readable to humans becomes more and more pressing.

Data visualization can facilitate meaningful analysis, accessibility, and interpretation of large datasets because it relies on human cognitive capabilities to process visual information (Gisbrecht 2015). Furthermore, it can support unanticipated discoveries by visually exploring and analyzing the data (Lowe and Matthee 2020). These discoveries could be a valuable resource in steering the creative process within architectural design. One reason for the necessity of data visualization in architecture is the potential to include acoustics in early design stages. Although architects already include performance as an early-on design driver for building components such as structure or facades, room acoustics are rarely included in the early design process (Badino, Shtrepi, and Astolfi 2020; Peters 2010). The reason for this is twofold. On the one hand, there is a lack of acoustic visualizations for architects and users with basic knowledge in acoustics. On the other hand, available visualizations are decoupled from the geometry that influences the sound. Nevertheless, geometry is essential for architects to understand the relationship between sound and geometry. This understanding would enable them to develop design workflows where they can manipulate the design and intuitively understand how this manipulation affects the room's acoustics.

Therefore, this research focuses on developing acoustic visualizations for big acoustic datasets for architects and

non-acoustic experts that can be integrated into the early design processes to enable acoustically informed design explorations.

BACKGROUND

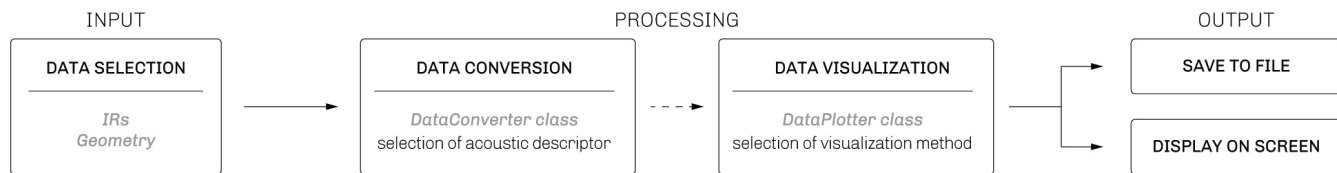
As big data becomes more prevalent in acoustic research, data visualizations have become increasingly important to interpret it. Lowe and Matthee describe that dimensionality reduction, interactivity, readability, and user assistance are key requirements of data visualization tools to interpret big data (Lowe and Matthee 2020). Building upon the key requirements introduced by them, we focus on four points to develop novel acoustic visualization for architectural applications: low- and high-dimensional visualizations, interactive visualizations, and usability of acoustic visualizations in architectural applications.

Low-dimensional Visualizations

From a data science perspective, an impulse response (IR) is a time series of float numbers that describe the sound's energy and phase over time. All acoustic parameters (called descriptors) that derive from it can be grouped into: (a) single value descriptors and (b) series-of-numbers descriptors. Single value descriptors include reverberation time (RT), clarity (C), strength (G), definition (D), center time (TS), and more. Series-of-numbers descriptors include frequency response and energy over time. Although acoustic analysis software such as Odeon and CATT-Acoustic (Odeon A/S 2022; CATT 2022) provides methods to calculate these acoustic descriptors, it requires expert knowledge in acoustics to decode their meaning.

High-dimensional Visualizations

High-dimensional data pose another challenge for accurate and readable visualization methods. Humans can only visually perceive three dimensions. Traditional data science visualization techniques, such as scatter plots and heat maps, can represent small or intermediate datasets in two or three dimensions. Although these visualizations are intuitive and may be used to identify bivariate correlations between variables, they require dimensionality reduction to arrange the data points in a lower-dimensional space. Dimensionality reduction methods such as Principal Component Analysis (Wold, Esbensen, and Geladi 1987) and *t*-distributed stochastic neighbor embedding (Van der Maaten and Hinton 2008) can compress attributes and reduce complexity. Although this compression is necessary to lower the dimensions down to two or three dimensions, it could lead to projection losses (Lowe and Matthee 2020). Projection loss describes a scenario where well-spread points in high-dimensional space appear falsely close in the low-dimensional projection (Lowe and Matthee 2020). Other limitations of dimensionality



2 Computational pipeline

reduction methods are that they treat the values as pure numbers, completely decoupled by the geometry that influenced them. This makes it especially difficult to include these methods in architectural applications as geometry is one of the key design components.

Interactive Visualizations

Interactive visualizations combine human and machine intelligence (Chen et al. 2019) to explore and uncover unexpected patterns in datasets (Cho et al. 2014). This visual analysis benefits “visual perception, interactive exploration, improved understanding, informed steering and intuitive interpretation” (Liu 2019). Furthermore, this type of analysis can have two approaches: a bottom-up approach that can potentially uncover patterns in the data (Ruan and Zhang 2017) and a top-down approach to test theories and search for evidence in the data (Genender-Feltheimer 2018; Mwangi, Soares, and Hasan 2014).

Usability of Acoustic Visualizations in Architectural Applications

During the design phase, architects explore various alternative design ideas. Early-stage design decisions have a significant impact on the final design’s quality and performance (Badino, Shtrepi, and Astolfi 2020). In contrast, late-stage design modifications can rarely compensate for poor early-stage choices. Therefore, it is essential to consider all factors early on to avoid potentially delaying the project, increasing the building cost, or impairing the overall design. To employ acoustic performance as a design driver, we must be able to quantify and interpret the acoustic effects of our geometric design choices. Currently, most of visual analytics are performed by acousticians equipped with the necessary knowledge and specialized acoustic analysis software. The reason for this is that this type of software is cumbersome to use and requires expert knowledge in acoustics. As a result, architects are discouraged from using them to evaluate their design, especially early on.

Commercially available acoustic analysis and visualization software are black boxes, providing insufficient feedback to the user and often no description of how the results were calculated (Kim et al. 2015). Furthermore, most commercial software do not provide APIs¹ or ways for external software to interface with them. This limitation, for example, hinders

form-finding studies using computational design because each design must be exported from the design software and imported into the acoustic analysis software for analysis. Therefore, acoustic visualization tools should be flexible for customization and easily expandable to accommodate study-specific requirements. The increased popularity of web-based python programming allows the development of visualization tools that are easily accessible and easy to use. Furthermore, their open-source nature allows users to customize them and extend their capabilities according to their needs.

This research proposes a novel approach for interactive visualizations of acoustic datasets for architects and non-acoustic experts. It introduces a series of simpler descriptors for users with basic knowledge in acoustics and describes methods for low- and high-dimensional data visualizations. It introduces visualization methods that incorporate the geometry that influenced the sound under study. The inclusion of the influential geometry provides a more detailed insight into the relationship between geometrical characteristics and the sound properties they influence. Moreover, this research proposes methods for interactive visualizations that allow users to explore the data from different angles and zoom closer to reveal more details. Interactivity also allows for animated content to display how sound properties change over time. Furthermore, it describes methods that allow the simultaneous display of multiple data points, enabling the user to compare data within the same visualization. Lastly, all visualizations are customizable and extendable, and the entire code is open-sourced and available at https://github.com/gramaziokohler/sdsc_data_driven_acoustic_design.

METHODS

Based on the topics described in the background, this section describes the computational workflow of the proposed visualization pipeline and demonstrates its use through a design scenario. For the design scenario, we used the open-source GIR Dataset (Xydis, Perraudin, Rust, Heutschi, et al. 2021). The dataset contains 920,712 real impulse responses (IRs) from 312 surfaces (2951 per surface). More details about how the dataset was collected can be found in (Rust et al. 2021).

Computational Workflow

The visualization pipeline is written in python and contains two main classes the DataConverter and the DataPlotter. The

TABLE 1 Available acoustic descriptors inside DataConverter class

Descriptor	Value type
Reverberation time	float
Clarity	float
Definition	float
Energy over time	list of floats
Cumulative energy over time	list of floats
Total cumulative energy	float
Frequency response	list of floats
Absorption coefficient	list of floats
Scattering coefficient	list of floats

DataConverter handles the data retrieval from the dataset and can convert the IR data to the desired acoustic descriptors, and the DataPlotter handles all the visualization computing. Open-source libraries are used to extend the core code. The *numpy* library handles the mathematical operation and the *scipy* library manages the audio-related computation such as Fast Fourier Transform (FFT) analysis and resampling. The visualization part of the code uses *seaborn* for computing heat maps, *matplotlib* for constructing all the static visualizations, and *pythreejs* for interactive visualizations. *Ipywidgets* is used to generate graphic widgets such as number sliders, button, and text inputs. The computational pipeline comprises of three main steps (see Figure 2). First, the input step where the data selection takes place, second, the processing step with data conversion and data visualization, and finally, the output step where the visualization is displayed or saved to a file.

Visualization Types

There are two types of visualizations possible with the computational workflow, explanatory visualizations, and comparative visualizations.

Explanatory Visualizations

The acoustic data visualization process begins with the selection of an input. For low-dimensional data the input refers to an IR measurement, in high dimensional visualization the input refers to several IRs. Afterwards the user selects the desired acoustic descriptor, and the DataConverter class returns the converted data. This data then can be passed to the DataPlotter for visualization. Table 1 shows all the available acoustic descriptors that DataConverter class can compute. Besides the standard output, each of these descriptors can also be normalized (from 0.0-1.0 or 0%-100% depending on the descriptor) or scaled using a scale factor. Table 2 shows all the available visualizations the DataPlotter class can generate. All visualizations have the option to display the output for the entire audio spectrum or per user-defined frequency bands (see Figure 5 and 6). This option, for example, allows the user to analyze the relationship between a geometrical design

TABLE 2 Available visualizations in DataPlotter class

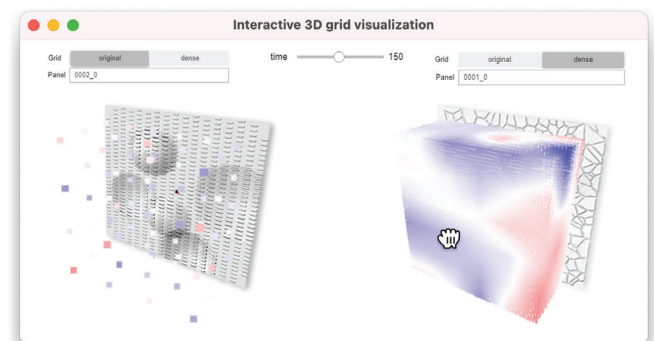
Visualization method	Interactive	Comparable
2D IR	no	yes
Energy over time	no	yes
Cumulative energy over time	no	yes
Frequency response	no	yes
Absorption coefficient	no	yes
Scattering coefficient	no	yes
2D grid	yes	yes
3D grid	yes	yes
3D polar	yes	yes

and groups of frequencies. This enables them to adjust their design to target specific frequencies. Finally, the output of the visualization can be either saved directly to a file, displayed on the screen, or both.

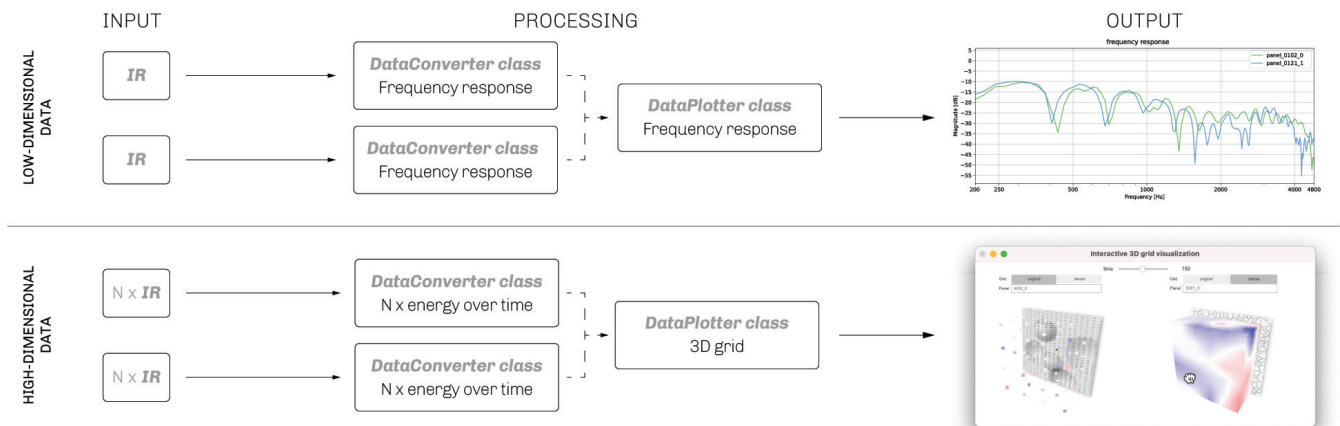
All visualizations follow the same principle. A sound is emitted at the source position, then the sound wave hits the room's surfaces and finally arrives at the receiver position. The visualizations show how these surfaces influenced the sound wave when the wave came in contact with them. Except for IR, Absorption coefficient, Scattering coefficient, and Frequency response, all visualizations can display absolute and relative values. For relative values, the values of a flat and smooth surface are taken as a baseline, and all values of the chosen surface relate to them. This way, the visualizations enable the evaluation of acoustic properties without professional assistance.

Comparative Visualizations

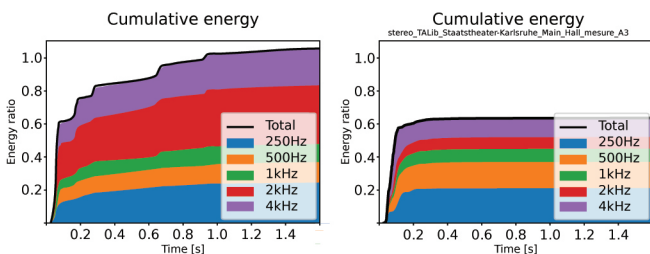
Several low- and high-dimensional descriptors can be used in comparative visualizations. The process is similar to the one described in the computational workflow section, except that for low-dimensional data (see Figure 4 top), users select



- Interactive comparative visualization of the energy over time: the left surface has the original density of the measuring grid and the right surface the denser interpolated grid; the time slider can be changed to show the energy for a specific time



4



5

- Computational workflow for comparative visualizations: to illustrate both output options, the low-dimensional visualization outputs a file and the high-dimensional visualization opens a window where users can interact with the data
- Cumulative energy over time visualization: (left) The current acoustic conditions of the scenario's room; the room is highly reverberant and has an unbalanced frequency response, and the 2kHz and 4kHz bands contain more energy than the other bands; (right) The Staatstheater in Karlsruhe as a reference of an acoustically treated space; black line represents the cumulative energy of the entire audio spectrum in the IR, and the colored sections for each filter band

two or more IRs and for high-dimensional data, two sets of multiple IRs. Then, they select the desired acoustic descriptor and generate the comparative visualization.

Interactivity

All high-dimensional data visualizations, explanatory, and comparative are displayed in a three-dimensional interactive window and include the surface geometry (Figure 3). They allow the user to pan, zoom, and rotate around the displayed data, breaking the barrier of a static data representation enabling users to study the relationship between geometry and sound intuitively. Furthermore, these visualizations support animated content. If a specific acoustic descriptor is selected, they can display the descriptor's value over time, adding an extra dimension to the visualization. When the comparative mode is active, the user can apply the same transformation to both visualizations and see the animated data simultaneously for both surfaces.

DESIGN SCENARIO

To show the entire workflow of the presented research, we are converting an existing room into an open-plan office space as an example. This section goes through several architectural design phases and demonstrates how the proposed visualizations enable a more acoustically informed design outcome. The design scenario has the architect as a user, and it consists of two main steps: (a) analysis and evaluation of the existing

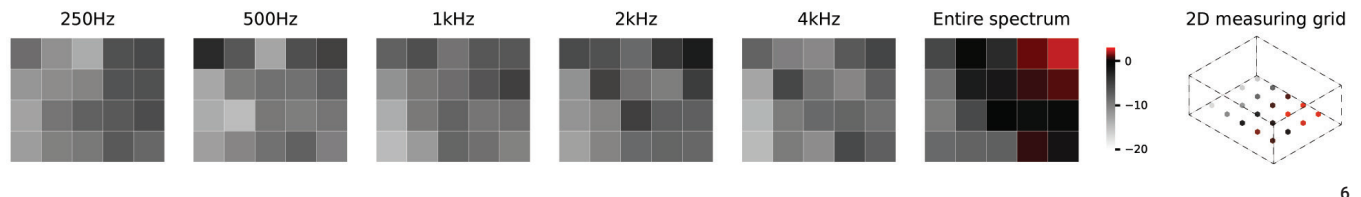
acoustic conditions and (b) exploration of acoustic design interventions.

Analysis and Evaluation

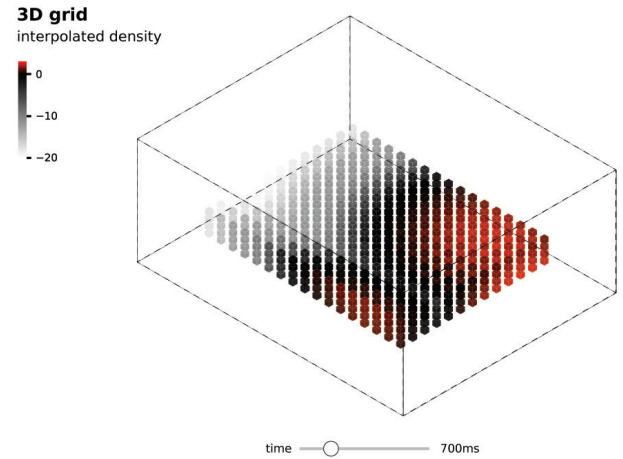
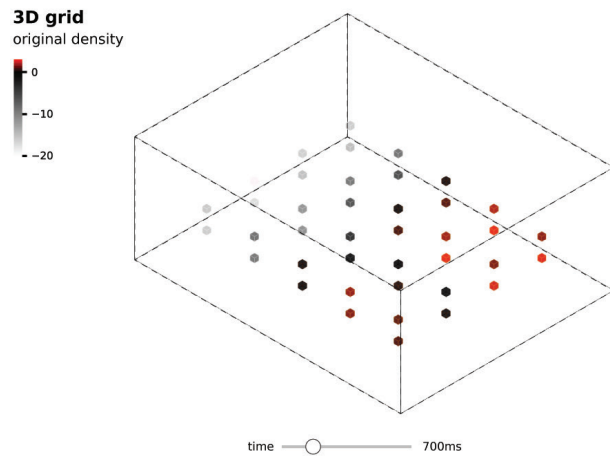
The process starts by analyzing the existing acoustic conditions of the room to understand the design actions necessary to improve the acoustic properties of the room for our open-plan office. After acquiring the impulse response measurements of the existing room, the user is ready to start the analysis. The computational workflow supports single IR measurements or multiple IR measurements recorded at several positions inside a room. Multiple IR measurements can be arranged in orthogonal two-dimensional or three-dimensional grids. Below we describe the three different arrangement options.

Single IR Measurements

Figure 5 shows explanatory two-dimensional visualizations of the Cumulative energy over time of single IRs. In this case, the user defines five filter bands, with a center frequency of 250 Hz, 500 Hz, 1 kHz, 2 kHz, and 4 kHz. The 250 Hz band is a low-pass filter (LPF), and the 4 kHz is a high-pass filter (HPF). The values are normalized from 0.0 to 1.0, with 0.0 representing no energy at the receiver position, and 1.0 that all the energy emitted by the source arrived at the receiver position. The Cumulative energy inside the room (see Figure 5 left) goes up quickly when the direct sound arrives, but



6



7

- 6 2D grid: total energy per filter band and for the entire audio spectrum; black represents energy values similar to the transmitted energy, and white 20dB less energy (equivalent to 100 times less energy); red squares are positions where the received energy is higher, indicating a local focusing effect
- 7 Interactive 3D grid visualization: (left) the original density grid with 40 measuring positions; (right) higher-density linear interpolated grid with 884 positions; in the interpolated grid the energy pattern is more visible

because the room is very reverberant due to its big volume and the lack of absorptive surfaces, the energy continues to rise for another 1.5 seconds. The small “steps” in the plot indicate when strong reflections arrive at the receiver position. These reflections can cause coloration, image shift, and flutter echoes, all negative characteristics for meeting rooms, lecture halls, and offices where speech clarity is important. Comparing it to the Cumulative energy of the Staatstheater in Karlsruhe,² we see that the energy rises quickly and then remains stable. This shows that there are no more reflections arriving at the receiver position. Furthermore, the total energy is lower because a portion of it (around 40%) was absorbed by the rooms surfaces. Rooms with Cumulative energies like Figure 5 (right) will have a clearer sound and be more relaxing working environments.

Multiple IR Measurements: 2D Measurement Grids

Multiple IR measurements of the same room enable a higher resolution analysis of the existing acoustic conditions. The visualization in Figure 6 can be used when the measurement positions are arranged in a two-dimensional grid. For this visualization we chose the total energy as a descriptor and each square represents one total energy value. This value

defines the total amount of sound energy that arrived at that location for the duration of the measurement. Looking at the entire spectrum grid we can clearly see the high sound energy concentrated at the corner of the room.

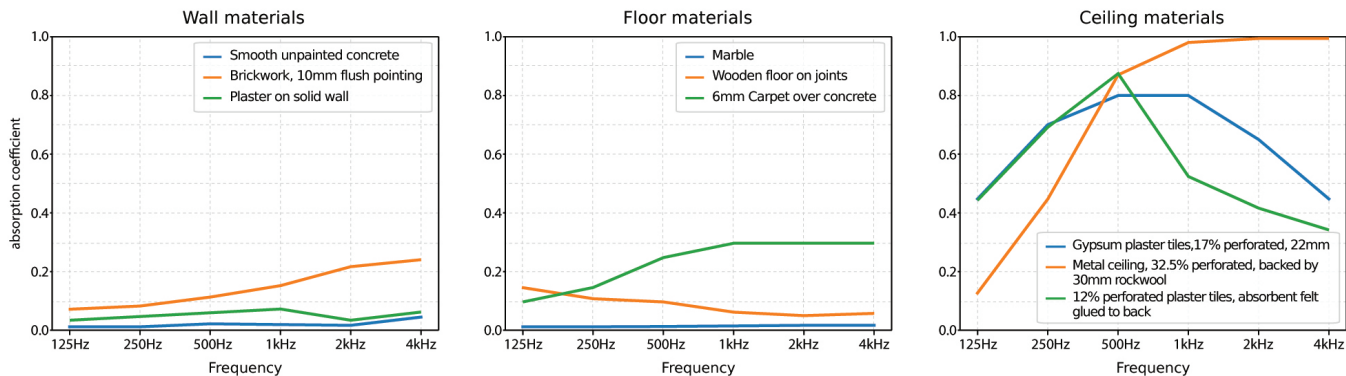
Multiple IR Measurements: 3D Measurement Grids

The visualization in Figure 7 can display values that were measured using a three-dimensional grid. The values are color-coded and displayed inside the room’s geometry. Users can pan and rotate the geometry to look at the values from a different angle or zoom in to take a closer look when the data is too dense. The grid’s density can be increased to make patterns more visible or decreased to reduce the visual complexity. Because this visualization also supports animated content, the energy values over time can be displayed by moving a time slider.

The analysis and evaluation show us clearly that the existing acoustic conditions are not suitable for a comfortable working environment. More precisely, some of the energy needs to be removed by means of absorptive materials. The ceiling and the floor are ideal locations for installing such materials. To balance the sound energy inside the room, limit disturbing flutter echoes, and prevent the room from becoming very unnatural from excess absorption, the walls can be designed in such a way that will diffuse sound. Below, we introduce different design explorations that the user can use to address the acoustic problems.

Design Explorations

This part of the scenario focuses on design interventions that



8 Comparative visualizations of absorption coefficients of different materials for the wall surfaces, the floor, and the ceiling of the room; values close to 0.0 represent no sound absorption, therefore, all energy is reflected, and values close to 1.0 complete absorption

can improve the acoustic conditions of the room. This part consists of two main steps, a) general acoustic treatment through proper surface material selection and b) design and adjustment of the wall's geometry to maximize the acoustic performance.

General Acoustic Treatment

The sound we hear is a combination of the direct sound coming straight from the source and indirect reflections from the surrounding surfaces. Sound hitting a surface is either transmitted, absorbed, or reflected; the ratio depends on the surface's acoustic properties. Based on the analysis of Figure 5, the user needs to lower the sound energy inside the room, emphasizing on the mid and high frequencies. This issue can be addressed by choosing appropriate materials that absorb the sound energy of mid- and high frequencies. Figure 8 shows three comparative two-dimensional visualization of Absorption coefficients for common architectural materials that can be used on the walls, floor, and ceiling of the room. The algorithm either receives the coefficients as a list of floats or computes them from IRs measured according the ISO 354:2003 standard (ISO 2003). In this case, the coefficient values for these visualizations are taken from a dataset that was created using data from www.acoustic.ua (Acoustic Traffic LLC 2022). Analyzing the three visualization reveals that a Brickwork would be preferable for the walls of the room (Figure 8 left, in orange), a thin carpet for the floor (Figure 8 middle, in green), and 32.5% perforated thin metal sheets, backed by 30 mm rock wool for the ceiling (Figure 8 right, in orange).

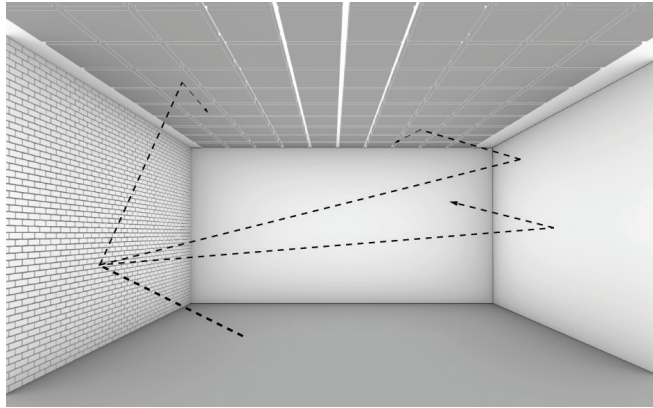
Design and Adjustment of Architectural Geometries

After deciding for brickwork as a wall material, the exact design of its geometry must be defined. Brickwork has a structured surface texture. Parts of the texture diffuse sound and other parts reflect sound in a specular way (Figure 9). The brickwork geometry can be optimized to redirect the reflected energy towards the ceiling (Figure 10). This way, more sound is reflected towards the ceiling where it gets absorbed.

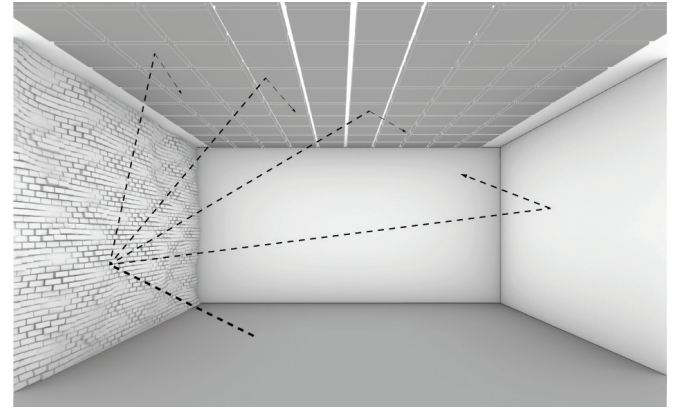
A recent research introduced a computational workflow to generate various acoustically informed diffusive surfaces, including brickworks (Xydis, Perraudin, Rust, Lytle, et al. 2021). The research uses self-organizing maps (SOM) to arrange sound diffusive architectural surfaces based on chosen acoustic properties. Using its design workflow, the SOM cell with the best matching unit contains two surfaces, a flat and a textured brickwork. To decide which of the two surfaces addresses better our acoustic requirements, the user can evaluate the direction of the reflected energy. Figures 11, 12, and 13 show interactive visualizations that display the direction and intensity of the reflected sound energy. The direction is represented as a ray leaving the surface and the intensity by the length and color of the line. Users can pan, zoom, or rotate to explore the sound directivity from different angles. Furthermore, the descriptors can be displayed for the entire audio spectrum or per user-defined frequency bands (see Figure 13). By comparing the two visualizations in Figure 11 and Figure 12 we can clearly see that the surface on the right redirects more energy upwards making it an ideal option for the design scenario.

RESULTS AND DISCUSSION

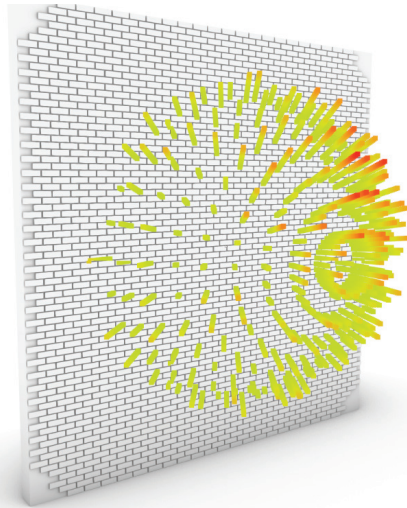
We have presented a novel approach to visualizing low and high-dimensional acoustic data. We described the computational workflow to produce these visualizations, its components, and how each of them contributes to the entire workflow. We demonstrated a series of visualizations and described how they could be used for acoustic studies. These visualizations show that thanks to their intuitive visual implementation, they can be used both by expert and non-expert users in acoustics. The proposed workflow addresses both experts and non-experts in acoustics users. Expert users can use standard and familiar acoustic descriptors and visualizations, while non-expert users are presented with a range of newly proposed and simplified descriptors and visualizations. Furthermore, both users benefit from the intuitive layout of the visualizations, especially from the interactivity of the



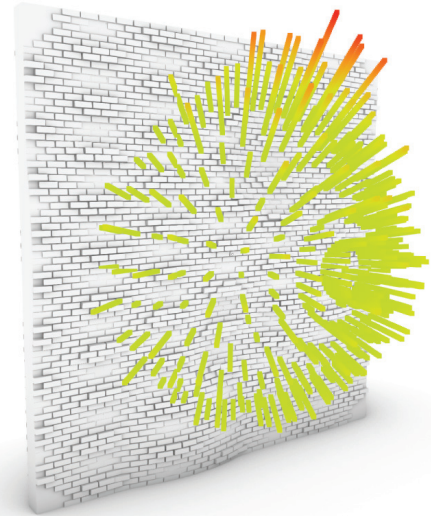
9



10



11

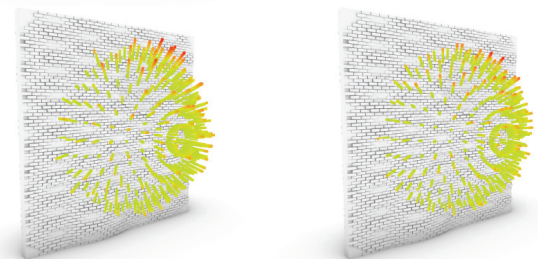


12

high-dimensional visualizations. Finally, including the geometry that influenced the sound in the visualizations allows for a deeper analysis of the mutual relationship between geometry and sound. We believe that these visualizations will help bring acoustics closer to the early phases of architectural design and enable a more integrative acoustic and architectural design exploration.

Limitations and Future Work

Despite the intuitive workflow, users still required basic knowledge of python to run the scripts. That is also true for extending and further customizing the visualizations. However, this visualization pipeline could be turned into a plugin for CAD software, eliminating the need for programming knowledge. In future steps, we are committed to continuing extending the visualization pipeline by adding more acoustic descriptors and visualization methods to it. Finally, we are confident that the proposed visualization methods will encourage researchers to create more open-source high-dimensional acoustic datasets.



13

- 9 The room from the design scenario with the flat brickwork. The black dashed lines illustrate the direction and intensity of the reflected sound according to the data from Figure 11. Most of the sound energy bounces back into the working area.
- 10 The room from the design scenario with the optimized brickwork. The black dashed lines illustrate the direction and intensity of the reflected sound according to the data from Figure 12. Here a significant portion of the sound energy gets reflected towards the absorbent ceiling.
- 11 3D polar sound directivity visualization of the total energy of the entire audio spectrum for the flat brickwork.
- 12 3D polar sound directivity visualization of the total energy of the entire audio spectrum for the textured brickwork
- 13 3D polar sound directivity visualization of the total energy of two frequency bands for the textured brickwork, left: 4kHz, right: 1kHz.

ACKNOWLEDGMENTS

This research steamed out of a collaborative and multidisciplinary project between Gramazio Kohler Research at ETH Zurich, the Swiss Data Science Center, the Laboratory for Acoustics/Noise Control at EMPA, and STRAUSS ELEKTROAKUSTIK GMBH. Therefore, the authors would like to thank Dr. Fernando Perez-Cruz, Dr. Nathanaël Perraudin, Dr. Kurt Heutschi, Kurt Eggenschwiler, and Jurgen Strauss for their inputs. Furthermore, we would like to thank Daniela Mitterberger for her valuable help and support preparing this paper.

NOTES

1. API stands for application programming interface, which is a set of definitions and protocols for building and integrating application software.
2. The IR is taken from the "Théâtre Acoustique Room Impulse Response Library," <https://www.lieuxperdus.com/convolver/download/>.

REFERENCES

- Acoustic Traffic LLC. "Absorption Coefficients." Accessed June 2, 2022. https://www.acoustic.ua/st/web_absorption_data_eng.pdf.
- Badino, Elena, Louena Shtrepi, and Arianna Astolfi. 2020. "Acoustic Performance-Based Design: A Brief Overview of the Opportunities and Limits in Current Practice." *Acoustics* 2 (2): 246–78.
- CATT. CATT-Acoustic. V.9.1. CATT. PC. 2022.
- Chen, Yi, Zeli Guan, Rong Zhang, Xiaomin Du, and Yunhai Wang. 2019. "A Survey on Visualization Approaches for Exploring Association Relationships in Graph Data." *Journal of Visualization* 22 (3): 625–39.
- Cho, Wonhee, Yoojin Lim, Hwangro Lee, Mohan Krishna Varma, Moonsoo Lee, and Eunmi Choi. 2014. "Big Data Analysis with Interactive Visualization Using R Packages." In *BigDataScience '14; Proceedings of the 2014 International Conference on Big Data Science and Computing*. New York, NY, USA: Association for Computing Machinery. 1–6.
- Genender-Feltheimer, Amy. 2018. "Visualizing High Dimensional and Big Data." *Procedia Computer Science, Cyber Physical Systems and Deep Learning Chicago* 140 (January): 112–21.
- Gisbrecht, Andrej. 2015. "Advances in Dissimilarity-Based Data Visualisation," PhD diss., Bielefeld University, Germany.
- ISO, International Organization for Standardization. 2003. *ISO 354:2003 Acoustics — Measurement of Sound Absorption in a Reverberation Room*. Geneva, Switzerland: ISO.
- Kim, Seokyeon, Seongmin Jeong, Sung Uk An, Jae Seok Yoo, Sang Min Han, Hanbyul Yeon, Sangbong Yoo, and Yun Jang. 2015. "Big Data Visual Analytics System for Disease Pattern Analysis." In *BigDAS '15; Proceedings of the 2015 International Conference on Big Data Applications and Services*. New York: Association for Computing Machinery. 175–79.
- Liu, Zhanping. 2019. "A Prototype Framework for Parallel Visualization of Large Flow Data." *Advances in Engineering Software* 130 (April): 14–23.
- Lowe, Joy, and Machdel Matthee. 2020. "Requirements of Data Visualisation Tools to Analyse Big Data: A Structured Literature Review." In *Responsible Design, Implementation and Use of Information and Communication Technology; Lecture Notes in Computer Science*, edited by M. Hattingh, M. Matthee, H. Smuts, I. Pappas, Y. K. Dwivedi, and M. Mäntymäki. Cham: Springer International Publishing. 469–80.
- Mwangi, Benson, Jair C. Soares, and Khader M. Hasan. 2014. "Visualization and Unsupervised Predictive Clustering of High-Dimensional Multimodal Neuroimaging Data." *Journal of Neuroscience Methods* 236 (October): 19–25.
- Odeon A/S. ODEON. V.17. Odeon A/S. PC. 2021.
- Peters, Brady. 2010. "Acoustic Performance as a Design Driver: Sound Simulation and Parametric Modeling Using SmartGeometry." *International Journal of Architectural Computing* 8 (3): 337–58.
- Ruan, Guangchen, and Hui Zhang. 2017. "Closed-Loop Big Data Analysis with Visualization and Scalable Computing." *Big Data Research, Tutorials on Tools and Methods using High Performance Computing resources for Big Data* 8 (July): 12–26.
- Rust, Romana, Achilleas Xydis, Kurt Heutschi, Nathanael Perraudin, Gonzalo Casas, Chaoyu Du, Jürgen Strauss, et al. 2021. "A Data Acquisition Setup for Data Driven Acoustic Design." *Building Acoustics* 28 (4). <https://doi.org/10.1177/1351010X20986901>.
- Sabine, Wallace Clement. 1922. *Collected Papers on Acoustics*. Cambridge: Harvard University Press.
- Van der Maaten, Laurens, and Geoffrey Hinton. 2008. "Visualizing Data Using T-SNE." *Journal of Machine Learning Research* 9 (11): 2579–2605.
- Van Long, Tran, and Lars Linsen. 2011. "Visualizing High Density Clusters in Multidimensional Data Using Optimized Star Coordinates." *Computational Statistics* 26 (4): 655.
- Wold, Svante, Kim Esbensen, and Paul Geladi. 1987. "Principal Component Analysis." *Chemometrics and Intelligent Laboratory Systems, Proceedings of the Multivariate Statistical Workshop for Geologists and Geochemists* 2 (1): 37–52.
- Xydis, Achilleas, Nathanaël Perraudin, Romana Rust, Kurt Heutschi, Gonzalo Casas, Oksana Riba Grognez, Kurt Eggenschwiler, Matthias Kohler, and Fernando Perez-Cruz. 2021. *GIR Dataset: A Geometry and Real Impulse Response Dataset*. V2. August 27, 2021. Distributed by Zenodo. <https://www.zenodo.org/record/5500519>.
- Xydis, Achilleas, Nathanaël Perraudin, Romana Rust, Beverly Ann Lytle, Fabio Gramazio, and Matthias Kohler. 2021. "Data-Driven Acoustic Design of Diffuse Surfaces Using Self-Organizing Maps." In *ACADIA 2021 Realignment: Toward Critical Computation; Proceedings of the 41st Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA)*, edited by K. Dorfler, S. Parascho, and J. Scott.

Achilleas Xydis is an architect and currently a doctoral researcher at the Chair of Architecture and Digital Fabrication (Gramazio Kohler Research) at ETH Zurich. He received his Diploma in Architecture from the University of Patras in 2010. In 2013, he completed the post-graduate Master of Advanced Studies in Architecture and Information at ETH Zurich. He focuses on combining machine learning techniques with architectural acoustics.

Chaoyu Du is a computational architect and doctoral researcher at the Block Research Group, ETH Zurich. She obtained her Master's degrees in architecture from Tongji University and urban design from the Technical University of Berlin, a MAS in Architecture and Digital Fabrication, and a CAS in Computer Science from ETH Zurich. Her research interests include computational structural design, complex fabrication, and machine intelligence in architecture.

Romana Rust is a computational architect and senior researcher at Gramazio Kohler Research, ETH Zurich within the Design++ initiative Centre for Augmented Computational Design in AEC. She is the co-coordinator of the Immersive Design Lab, a lab for collaborative research and teaching in the field of extended reality and machine learning in architecture and construction. Her particular interest is the development of innovative computational methods that integrate multiple design objectives such as geometry, acoustics, materiality, and robotic fabrication.

Fabio Gramazio and Matthias Kohler are professors of Architecture and Digital Fabrication at ETH Zurich. In 2000, they founded the architecture practice Gramazio & Kohler, which realized numerous award-winning projects. Opening the world's first architectural robotic laboratory at ETH Zurich, Gramazio & Kohler's research has been formative in the field of digital architecture, setting precedence and de facto creating a new research field merging advanced architectural design and additive fabrication processes through the customized use of industrial robots.